HW0331

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5.5

(a.)(b.)(c.)

a.

$x_t = t$

$$Var(b_2|X) = \frac{\sigma^2}{\Sigma(t-\bar{t})^2}$$

$$= \frac{\sigma^2}{\Sigma t^2 - \frac{(\Sigma t)^2}{T}}$$

$$= \frac{\sigma^2}{\frac{T(T+1)(2T+1)}{6} - \frac{\left(\frac{T(T+1)}{2}\right)^2}{T}}$$

$$= \frac{\sigma^2}{\frac{2T(T+1)(2T+1) - 3T(T+1)(T+1)}{12}}$$

$$= \frac{\sigma^2}{\frac{T(T+1)(T-1)}{12}}$$

$$= \frac{12\sigma^2}{T(T^2-1)}$$

$$\lim_{T\to\infty} var(b_2|X) = \lim_{T\to\infty} \frac{12\sigma^2}{T^3(1-\frac{1}{T^2})} = 0$$

b_2 is consistent. ※

b.

$$x_t = (0.5)^t$$

$$Var(b_2|x) = \frac{\sigma^2}{\Sigma(x_i-\bar{x})^2} = \frac{\sigma^2}{\Sigma x_i^2 - N\bar{x}^2}$$

$$= \frac{\sigma^2}{\Sigma((0.5)^t)^2 - T\times\left(\frac{\Sigma(0.5)^t}{T}\right)^2}$$

$$= \frac{\sigma^2}{\frac{0.25(1-0.25^T)}{1-0.25} - T\times\left(\frac{\frac{0.5(1-0.5^T)}{1-0.5}}{T}\right)^2}$$

$$= \frac{\sigma^2}{\frac{(1-0.25^T)}{3} - \frac{(1-0.5^T)^2}{T}}$$

$$\lim_{T\to\infty} Var(b_2|x) = \frac{\sigma^2}{\frac{1}{3}-0} = 3\sigma^2 \neq 0$$

b_2 is not consistent. #

c.

(a) T持續增加提供更多 x_t 並確立 x, y 之間的關係 (β_2)

(b)

T持續增加讓 x_t 最後趨近於 0 便不能提供更多關於 β_2 的資訊。

5.6

(a.)(b.)(c.)

a.

$H_0: \beta_2 = 0 \quad H_1: \beta_2 \neq 0$

$$t^* = \frac{b_2 - 0}{se(b_2)} = \frac{3}{\sqrt{4}} = 1.5$$

拒絕 H_0 if $|t^*| > t_{60, 0.025} = 2$

$\because |t^*| = 1.5 < 2 \quad \therefore$ 不拒絕 H_0

b.

$H_0: \beta_1 + 2\beta_2 = 5 \quad H_1: \beta_1 + 2\beta_2 \neq 5$

$$t^* = \frac{(b_1 + 2b_2) - 5}{se(b_1 + 2b_2)} = \frac{(2 + 2\times 3) - 5}{\sqrt{3 + 2^2 \times 4 + 2\times 1\times 2\times(-2)}} = \frac{3}{\sqrt{11}}$$

拒絕 H_0 if $|t^*| > t_{60, 0.025} = 2$

$\because |t^*| = 0.9045 < 2 \quad \therefore$ 不拒絕 H_0

c.

$H_0: \beta_1 - \beta_2 + \beta_3 = 4 \quad H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$

$$t^* = \frac{(b_1 - b_2 + b_3) - 4}{se(b_1 - b_2 + b_3)} = \frac{(2 - 3 + (-1)) - 4}{\sqrt{3 + (-1)^2 \times 4 + 3 + 2\times 1\times(-1)\times(-2) + 2\times 1\times 1\times 1 + 2\times 1\times(-1)\times 0}}$$

$$= \frac{-6}{4} = -1.5$$

拒絕 H_0 if $|t^*| > t_{60, 0.025} = 2$

$\because |t^*| = 1.5 < 2 \quad \therefore$ 不拒絕 H_0

5.20

(a.)(b.)

a.

$$\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1$$

$$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$$

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

$$= \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$E(\hat{\beta}_{2,mean} \mid X) = \beta_2 + \frac{\overset{0}{\overset{\shortparallel}{E(\bar{e}_2 \mid X)}} - \overset{0}{\overset{\shortparallel}{E(\bar{e}_1 \mid X)}}}{\bar{X}_2 - \bar{X}_1} = \beta_2$$

$$E(\hat{\beta}_{2,mean}) = E\left(E(\hat{\beta}_{2,mean} \mid X)\right) = E(\beta_2) = \beta_2$$

因此，$\hat{\beta}_{2,mean}$ 是 β_2 的不偏 estimator

b.

$$Var(\hat{\beta}_{2,mean} \mid X) = Var\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{X}_2 - \bar{X}_1} \mid X\right) = \frac{1}{(\bar{X}_2 - \bar{X}_1)^2} \times Var(\bar{e}_2 - \bar{e}_1 \mid X)$$

$$Var(\bar{e}_1 \mid X) = Var\left(\frac{1}{3}\sum_{i=1}^{3} e_i \mid X\right) = \frac{1}{3^2}\sum_{i=1}^{3} Var(e_i \mid X) = \frac{1}{3^2} \times 3 \times \sigma^2$$

$$= \frac{\sigma^2}{3}$$

$Var(\bar{e}_2 \mid X) = \frac{\sigma^2}{3}$ 同理

$$Var(\bar{e}_2 - \bar{e}_1 \mid X) = \frac{\sigma^2}{3} + \frac{\sigma^2}{3} = \frac{2}{3}\sigma^2$$

代回原式

$$Var(\hat{\beta}_{2,mean} \mid X) = \frac{\frac{2}{3}\sigma^2}{(\bar{X}_2 - \bar{X}_1)^2} \quad ※$$

(c.)(d.)

c.

$$Var(\hat{\beta}_{2,mean}) = E[Var(\hat{\beta}_{2,mean} \mid X)] + Var(E(\hat{\beta}_{2,mean} \mid X))$$

$$Var(E(\hat{\beta}_{2,mean} \mid X)) = Var(\beta_2) = 0$$

$$E[Var(\hat{\beta}_{2,mean} \mid X)] = \frac{2}{3}\sigma^2 E\left(\frac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right)$$

代回原式

$$Var(\hat{\beta}_{2,mean}) = \frac{2}{3}\sigma^2 E\left(\frac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right)$$

d.

如果要 consistent，當 $N \to \infty$，$Var(\hat{\beta}_{2,mean}) \to 0$

$$Var(\hat{\beta}_{2,mean}) = \frac{4}{N}\sigma^2 E\left(\frac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right)$$

$$\lim_{N \to \infty} \frac{4}{N}\sigma^2 E\left(\frac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right) = 0 \quad \#$$

(e.)(f.)

	N	var_b2_given_x	var_b2mean_given_x	inv_sx2	four_over_gap2
1	100	1.24362517	1.6631963	0.1243625	0.1663196
2	500	0.24402399	0.3260404	0.1220120	0.1630202
3	1000	0.12290628	0.1656282	0.1229063	0.1656282
4	5000	0.02413174	0.0323161	0.1206587	0.1615805

f.

① 根據結果 (e) $\begin{cases} Var(b_2|X) = \dfrac{\sigma^2}{N s_x^2}, \; Var(\hat{\beta}_{2,mean}|X) = \dfrac{4\sigma^2}{N(\bar{x}_2-\bar{x}_1)^2} \\ E[(s_x^2)^{-1}] \doteqdot 0.12 \quad E\left[\dfrac{4}{(\bar{x}_2-\bar{x}_1)^2}\right] \doteqdot 0.16 \end{cases}$

因此 $Var(b_2|X) \approx \dfrac{0.12 \times 1{,}000}{N}$

$Var(\hat{\beta}_{2,mean}|X) \approx \dfrac{0.16 \times 1{,}000}{N}$

$Var(b_2|X) < Var(\hat{\beta}_{2,mean}|X)$

② 當N增加，兩者分別往 0.12 和 0.16 靠近

③ 因 $x_i \sim U(0,10)$

則當 $N \to \infty$, $\bar{x}_1 \xrightarrow{p} 2.5$ $\bar{x}_2 \xrightarrow{p} 7.5$

$Var(\hat{\beta}_{2,mean}|X) \xrightarrow{p} \dfrac{4\sigma^2}{N \times (7.5-2.5)^2} \to 0$ #

(g.)(h.)

g.

$$E\left[(S_x^2)^{-1}\right] \xrightarrow{p} 0.12$$

$$S_x^2 \xrightarrow{p} Var(X) = \frac{(10-0)^2}{12}$$

$$(S_x^2)^{-1} \xrightarrow{p} \frac{12}{100} = 0.12 \ \#$$

$$E\left[\frac{4}{(\bar{X}_2-\bar{X}_1)^2}\right] \xrightarrow{p} 0.16$$

前半段 $U(0,5)$ 後半段 $U(5,10)$

$\bar{X}_1 \xrightarrow{p} 2.5$ $\bar{X}_2 \xrightarrow{p} 7.5$

$$\bar{X}_2-\bar{X}_1 \xrightarrow{p} 7.5-2.5=5$$

$$\frac{4}{(\bar{X}_2-\bar{X}_1)^2} \xrightarrow{p} \frac{4}{5^2} = 0.16 \ \#$$

h.

若非 increasing 而是 random，$\bar{X}_1 \xrightarrow{p} 5$ $\bar{X}_2 \xrightarrow{p} 5$

$$(\bar{X}_2-\bar{X}_1) \xrightarrow{p} 0$$

則 $Var(\hat{\beta}_{2,mean} \mid X) = \dfrac{4\sigma^2}{N(\bar{X}_2-\bar{X}_1)^2}$

無法收斂到 0（可能會無限大）

故 $\hat{\beta}_{2,mean}$ would not be consistent.